Name : Preity Mestri TE - B3 batch Roll No. 13256 Practical No:3 Statistical Summary . Types of Variables.

In [3]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
```

In [5]:
```
df=pd.read_csv("C:/Users/Preity/Downloads/Placement_Data_Full_Class.csv")
```

In [6]:
```
df
```

Out[6]:

	sl_no	gender	ssc_p	ssc_b	hsc_p	hsc_b	hsc_s	degree_p	degree_t
0	1	M	67.00	Others	91.00	Others	Commerce	58.00	Sci&Tech
1	2	M	79.33	Central	78.33	Others	Science	77.48	Sci&Tech
2	3	M	65.00	Central	68.00	Central	Arts	64.00	Comm&Mgmt
3	4	M	56.00	Central	52.00	Central	Science	52.00	Sci&Tech
4	5	M	85.80	Central	73.60	Central	Commerce	73.30	Comm&Mgmt
...	...	...	...	...	...	...	...	...	...
210	211	M	80.60	Others	82.00	Others	Commerce	77.60	Comm&Mgmt
211	212	M	58.00	Others	60.00	Others	Science	72.00	Sci&Tech
212	213	M	67.00	Others	67.00	Others	Commerce	73.00	Comm&Mgmt
213	214	F	74.00	Others	66.00	Others	Commerce	58.00	Comm&Mgmt
214	215	M	62.00	Central	58.00	Others	Science	53.00	Comm&Mgmt

215 rows × 15 columns

In [9]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 215 entries, 0 to 214
Data columns (total 15 columns):
 #   Column          Non-Null Count  Dtype  
---  ------          --------------  -----  
 0   sl_no           215 non-null    int64  
 1   gender          215 non-null    object 
 2   ssc_p           215 non-null    float64
 3   ssc_b           215 non-null    object 
 4   hsc_p           215 non-null    float64
 5   hsc_b           215 non-null    object 
 6   hsc_s           215 non-null    object 
 7   degree_p        215 non-null    float64
 8   degree_t        215 non-null    object 
 9   workex          215 non-null    object 
 10  etest_p         215 non-null    float64
 11  specialisation  215 non-null    object 
 12  mba_p           215 non-null    float64
 13  status          215 non-null    object 
 14  salary          148 non-null    float64
dtypes: float64(6), int64(1), object(8)
memory usage: 25.3+ KB
```

In [10]:
```
df.shape
```

Out[10]:
```
(215, 15)
```

In [11]:
```
df.describe
```

Out[11]:
```
<bound method NDFrame.describe of      sl_no gender  ssc_p    ssc_b  hsc_p    h
sc_b     hsc_s  degree_p  \
0         1      M  67.00   Others  91.00   Others  Commerce     58.00
1         2      M  79.33  Central  78.33   Others   Science     77.48
2         3      M  65.00  Central  68.00  Central      Arts     64.00
3         4      M  56.00  Central  52.00  Central   Science     52.00
4         5      M  85.80  Central  73.60  Central  Commerce     73.30
..      ...    ...    ...      ...    ...      ...       ...       ...
210     211      M  80.60   Others  82.00   Others  Commerce     77.60
211     212      M  58.00   Others  60.00   Others   Science     72.00
212     213      M  67.00   Others  67.00   Others  Commerce     73.00
213     214      F  74.00   Others  66.00   Others  Commerce     58.00
214     215      M  62.00  Central  58.00   Others   Science     53.00

      degree_t workex  etest_p specialisation  mba_p      status    salary
0     Sci&Tech     No     55.0         Mkt&HR  58.80      Placed  270000.0
1     Sci&Tech    Yes     86.5        Mkt&Fin  66.28      Placed  200000.0
2    Comm&Mgmt     No     75.0        Mkt&Fin  57.80      Placed  250000.0
3     Sci&Tech     No     66.0         Mkt&HR  59.43  Not Placed       NaN
4    Comm&Mgmt     No     96.8        Mkt&Fin  55.50      Placed  425000.0
..         ...    ...      ...            ...    ...         ...       ...
210  Comm&Mgmt     No     91.0        Mkt&Fin  74.49      Placed  400000.0
211   Sci&Tech     No     74.0        Mkt&Fin  53.62      Placed  275000.0
212  Comm&Mgmt    Yes     59.0        Mkt&Fin  69.72      Placed  295000.0
213  Comm&Mgmt     No     70.0         Mkt&HR  60.23      Placed  204000.0
214  Comm&Mgmt     No     89.0         Mkt&HR  60.22  Not Placed       NaN

[215 rows x 15 columns]>
```

In [12]:
```
df.isnull().sum()
```

Out[12]:
```
sl_no             0
gender            0
ssc_p             0
ssc_b             0
hsc_p             0
hsc_b             0
hsc_s             0
degree_p          0
degree_t          0
workex            0
etest_p           0
specialisation    0
mba_p             0
status            0
salary           67
dtype: int64
```

In [13]:
```
df['salary'].mean()
```

Out[13]:
```
np.float64(288655.4054054054)
```

In [14]:
```
df['ssc_p'].median()
```

Out[14]:
```
67.0
```

In [15]:
```
df['salary'] = df['salary'].fillna(df['salary'].mean())
```

In [16]:
```
df.head()
```

Out[16]:

	sl_no	gender	ssc_p	ssc_b	hsc_p	hsc_b	hsc_s	degree_p	degree_t	w
0	1	M	67.00	Others	91.00	Others	Commerce	58.00	Sci&Tech	
1	2	M	79.33	Central	78.33	Others	Science	77.48	Sci&Tech	
2	3	M	65.00	Central	68.00	Central	Arts	64.00	Comm&Mgmt	
3	4	M	56.00	Central	52.00	Central	Science	52.00	Sci&Tech	
4	5	M	85.80	Central	73.60	Central	Commerce	73.30	Comm&Mgmt	

In [17]:
```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df['gender'] = le.fit_transform(df['gender'])
```

In [18]:
```
df['status'] = le.fit_transform(df['status'])
```

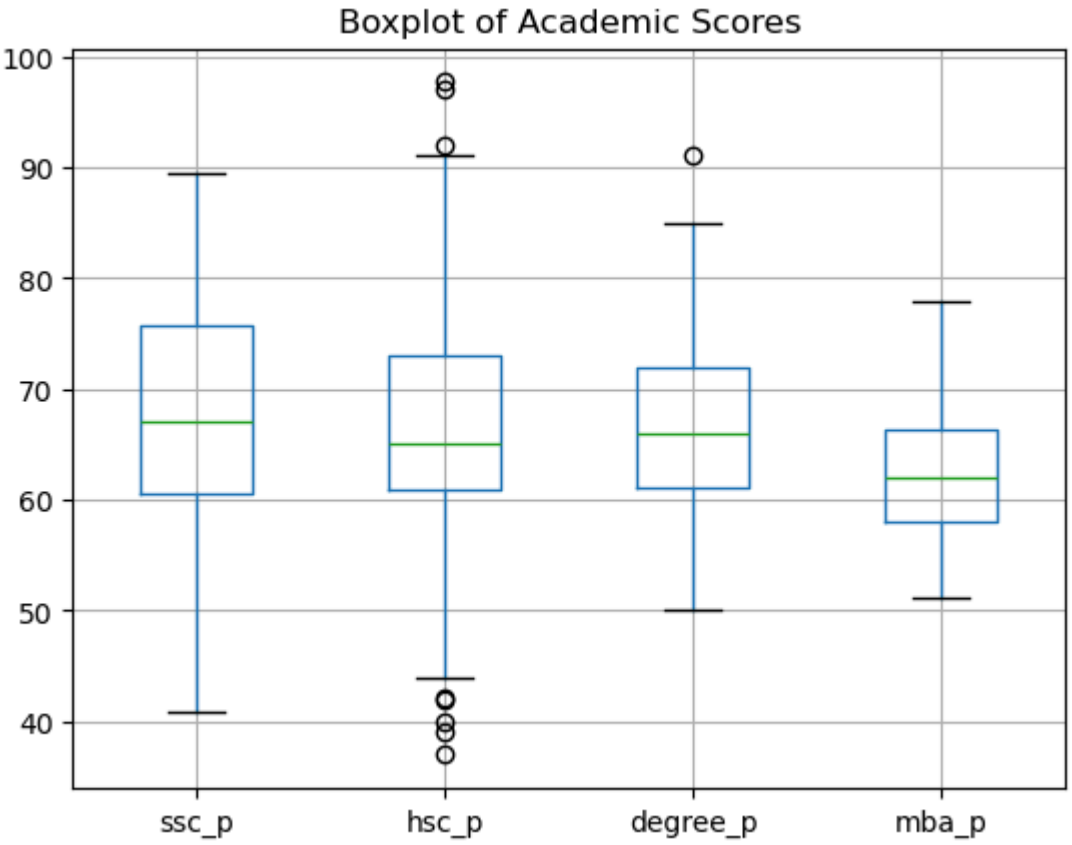
In [19]:
```
df['workex'] = le.fit_transform(df['workex'])
```

In [20]:
```
df['specialisation'] = le.fit_transform(df['specialisation'])
df.head()
```

Out[20]:

	sl_no	gender	ssc_p	ssc_b	hsc_p	hsc_b	hsc_s	degree_p	degree_t	w
0	1	1	67.00	Others	91.00	Others	Commerce	58.00	Sci&Tech	
1	2	1	79.33	Central	78.33	Others	Science	77.48	Sci&Tech	
2	3	1	65.00	Central	68.00	Central	Arts	64.00	Comm&Mgmt	
3	4	1	56.00	Central	52.00	Central	Science	52.00	Sci&Tech	
4	5	1	85.80	Central	73.60	Central	Commerce	73.30	Comm&Mgmt	

In [21]:
```
df.boxplot(column=['ssc_p','hsc_p','degree_p','mba_p'])
plt.title("Boxplot of Academic Scores")
plt.show()
```



In [22]:
```
z = np.abs(stats.zscore(df['mba_p']))
df[z > 3]
```

Out[22]:

	sl_no	gender	ssc_p	ssc_b	hsc_p	hsc_b	hsc_s	degree_p	degree_t	workex	etest_p

In [23]:
```
Q1 = df['hsc_p'].quantile(0.25)
Q3 = df['hsc_p'].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
print("Lower Bound:", lower)
print("Upper Bound:", upper)
```

```
Lower Bound: 42.75
Upper Bound: 91.15
```

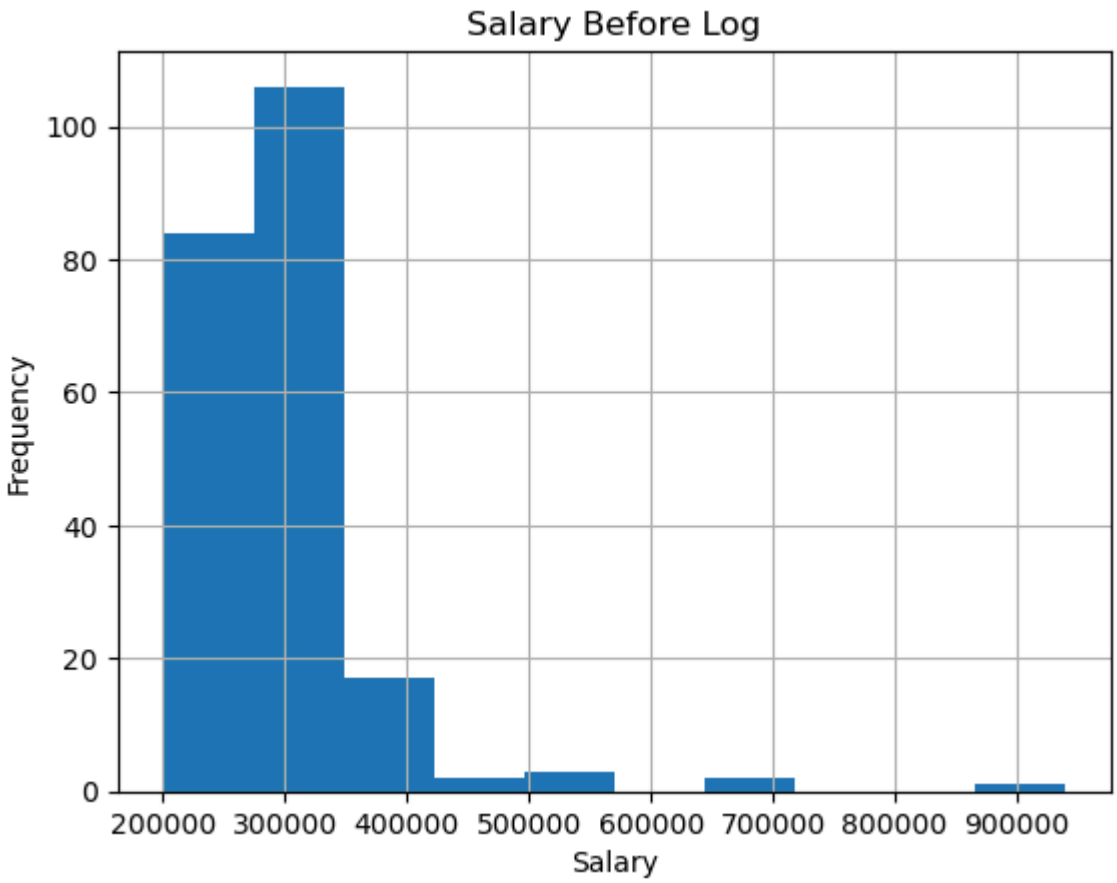
In [24]:
```
df[(df['hsc_p'] < lower) | (df['hsc_p'] > upper)]
```

Out[24]:

	sl_no	gender	ssc_p	ssc_b	hsc_p	hsc_b	hsc_s	degree_p	degree_t
24	25	1	76.50	Others	97.70	Others	Science	78.86	Sci&Tech
42	43	1	49.00	Others	39.00	Central	Science	65.00	Others
49	50	0	50.00	Others	37.00	Others	Arts	52.00	Others
120	121	1	58.00	Others	40.00	Others	Science	59.00	Comm&Mgmt
134	135	0	77.44	Central	92.00	Others	Commerce	72.00	Comm&Mgmt
169	170	1	59.96	Others	42.16	Others	Science	61.26	Sci&Tech
177	178	0	73.00	Central	97.00	Others	Commerce	79.00	Comm&Mgmt
206	207	1	41.00	Central	42.00	Central	Science	60.00	Comm&Mgmt

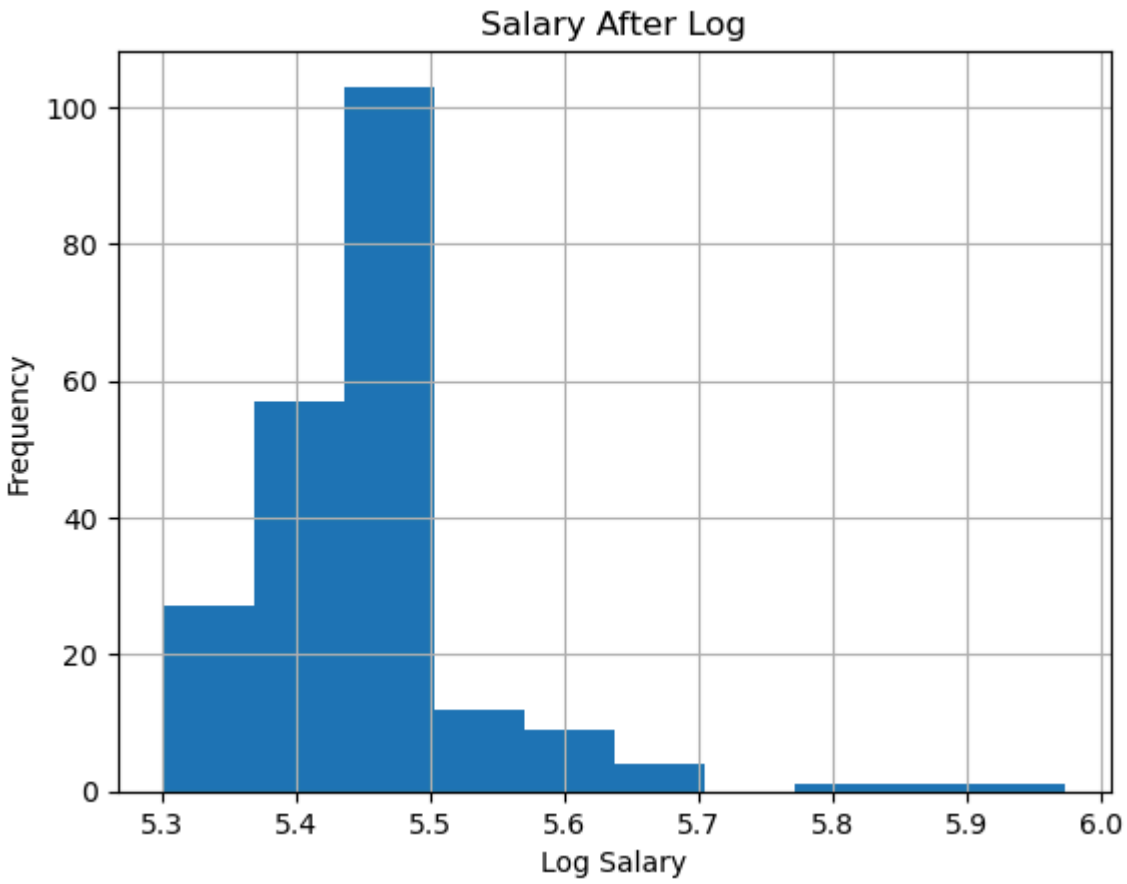
In [25]:
```
median = df['hsc_p'].median()
df['hsc_p'] = np.where((df['hsc_p'] < lower) | (df['hsc_p'] > upper),median,df['
```

In [26]:
```
df['salary'].hist()
plt.title("Salary Before Log")
plt.xlabel("Salary")
plt.ylabel("Frequency")
plt.show()
df['salary'].skew()
```



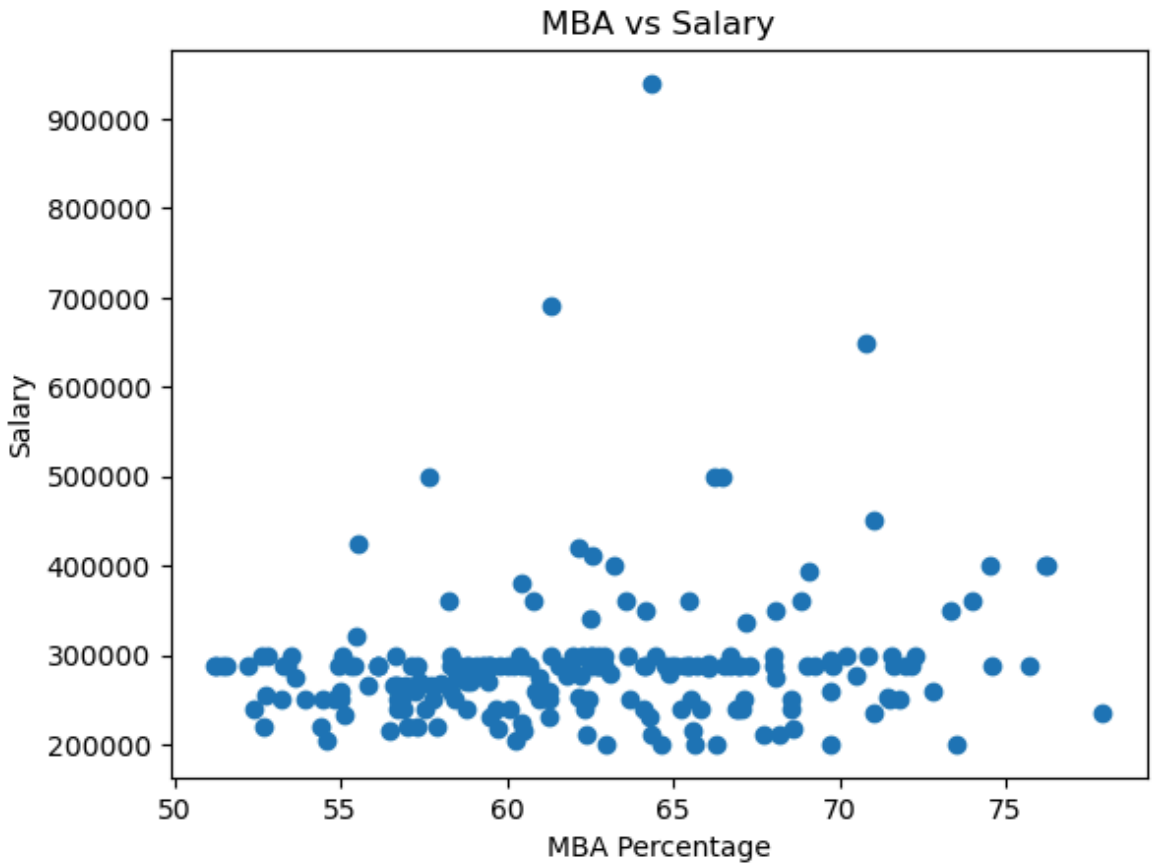
Out[26]: np.float64(4.288798850070048)

In [27]:
```
df['log_salary'] = np.log10(df['salary'])
df['log_salary'].hist()
plt.title("Salary After Log")
plt.xlabel("Log Salary")
plt.ylabel("Frequency")
plt.show()
df['log_salary'].skew()
```



Out[27]: np.float64(1.9500125921488516)

In [28]:
```
plt.scatter(df['mba_p'], df['salary'])
plt.xlabel("MBA Percentage")
plt.ylabel("Salary")
plt.title("MBA vs Salary")
plt.show()
```



In [29]:
```
mad = (df['salary'] - df['salary'].mean()).abs().mean()
print("Mean Absolute Deviation:", mad)
```

```
Mean Absolute Deviation: 39441.1062225016
```

In [30]:
```
variance = df['salary'].var()
print("Variance:", variance)
```

```
Variance: 5999726288.204086
```

In [31]:
```
std_dev = df['salary'].std()
print("Standard Deviation:", std_dev)
```

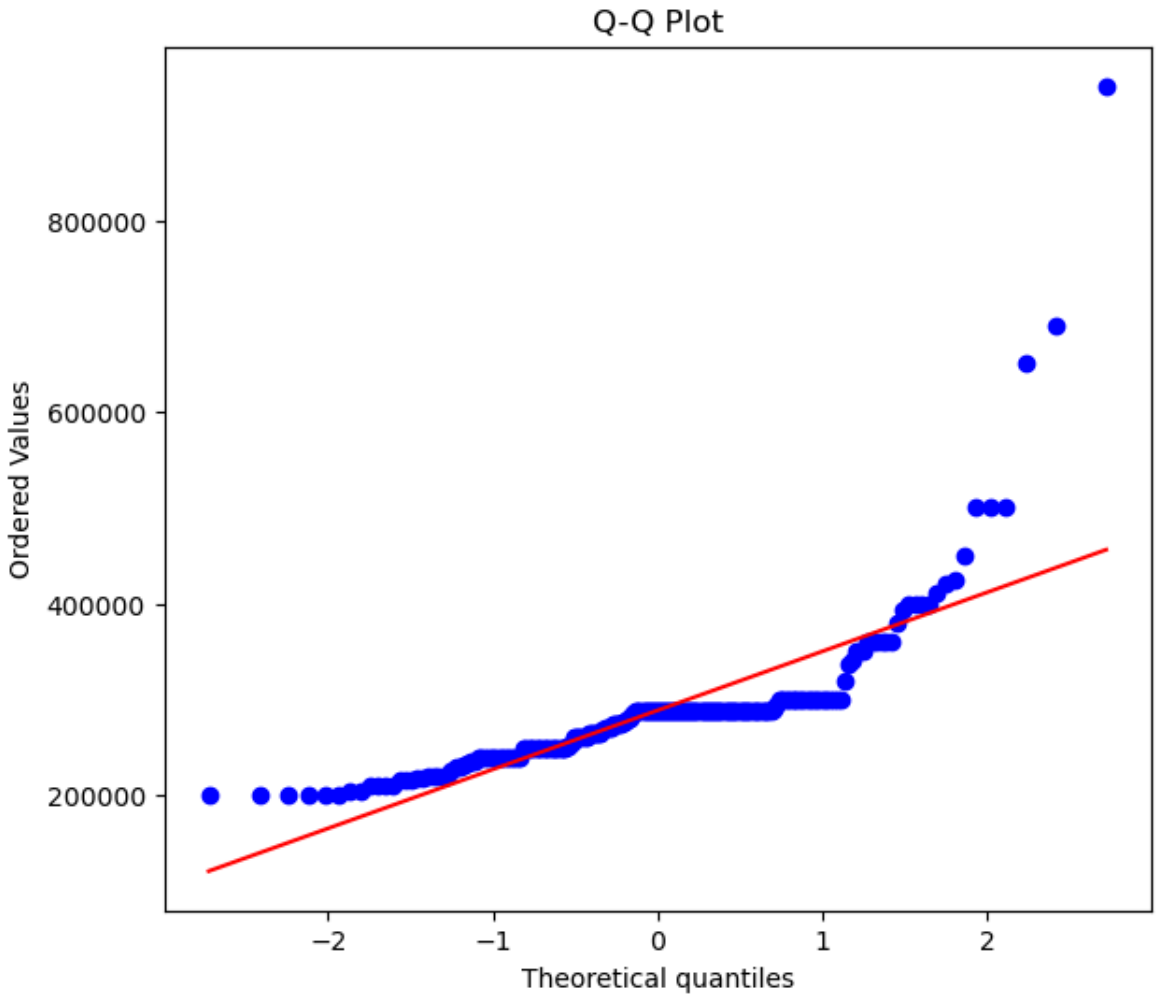
```
Standard Deviation: 77457.90010195272
```

In [32]:
```
data_range = df['salary'].max() - df['salary'].min()
print("Range:", data_range)
```

```
Range: 740000.0
```

In [33]:
```
def dgraph(df, variable):
    plt.figure(figsize=(15,6))
    plt.subplot(1, 2, 1)
    stats.probplot(df[variable], dist='norm', plot=plt)
    plt.title("Q-Q Plot")
    plt.show()
```

In [34]:
```
dgraph(df, 'salary')
```


Q-Q Plot
Ordered Values
800000
600000
400000
200000
−2
−1
0
1
2
Theoretical quantiles

In [ ]: